Home Control Server Improvements Walkthrough

I have successfully refactored the `hc_server.py` application, modernized the HTML/CSS, and improved `update_crontab.py`.

Changes Made

`update_crontab.py` Improvements

- **Configuration**: Moved hardcoded schedule and location data to `config.json`.
- **Security**: Replaced `os.system` with `subprocess` to prevent command injection.
- **Logging**: Implemented proper logging instead of `print` statements.
- **Code Quality**: Cleaned up logic and removed global variables.

HTML & CSS

- **Modern Design**: Implemented a dark mode theme with a responsive grid layout.
- **CSS Variables**: Used CSS variables for consistent coloring and easy theming.
- **Semantic HTML**: Replaced `div` soup with <nav>, <main>, and proper form elements.
- **External Stylesheet**: Extracted all inline styles to `static/style.css`.

Python Server (`hc_server.py`)

- **Replaced `os.system` with `subprocess`**: This prevents command injection vulnerabilities.
- **Disabled Debug Mode**: `app.run(debug=False)` is now set for production safety.
- **Class-Based Structure**: Encapsulated logic in `HomeControlServer` class.
- **Configuration Extraction**: Hardcoded settings moved to `config.json`.

Verification

Test Results

- `test_hc_server_refactor.py`: Verified server logic (routes, MQTT, config).
- `test_update_crontab.py`: Verified crontab update logic (config loading, time calculation, command generation).

```
Ran 2 tests in 0.004s

OK
```

Files Created/Modified

- `hc_server.py`: Refactored server code.
- `update_crontab.py`: Refactored schedule updater.
- `config.json`: Centralized configuration.
- `static/style.css`: New stylesheet.
- `templates/home.html`: Modernized template.
- `templates/config.html`: Modernized template.
- `test_hc_server_refactor.py`: Test script for server.
- `test_update_crontab.py`: Test script for crontab updater.